Computer Integrated Manufacturing

Additive Manufacturing of Off-road Tire for Autonomous Rover

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Table of Contents

1. Introduction 3
2. Analysis of Manufacturing Constraints and Application of DfX Principles 3
2.1 Selected DfX Principles for FDM 3
2.2 Additive-Manufacturing Process-Chain Guidelines 4
3.Optimization of the Component for Manufacturability. 5
3.1 Solutions for Manufacturability 6
3.2 Off-Road Optimization 6
3.3 Integration in Siemens NX 7
4. Creation of the Optimized Component for Additive Manufacturing 7
4.1 Exporting from Siemens NX 7
4.2 PrusaSlicer print parameters 8
5. Conclusion 9
6. Bibliography 9

1. Introduction

Off-road vehicle tires must strike a balance between traction, compliance, and durability when driving on loose, uneven terrains like mud, gravel, and sand. This project optimizes an existing tire model for dual-material free-form production on the Original Prusa XL (5T tool multi-extruder) utilizing Fused Deposition Modeling (FDM).The revised tire provides improved grip, less ground sinkage, and prolonged service life by combining a soft, energy absorbing inner core (TPU 95A) with a rigid, wear-resistant outer tread (PETG) that can be manufactured on a desktop FDM machine.

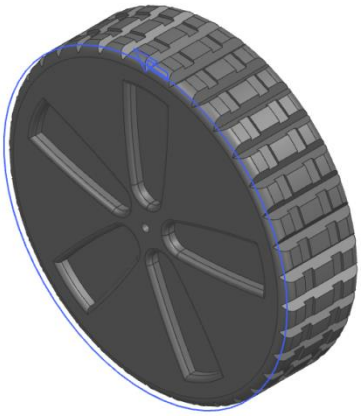


Figure 1.1 The Original Rover wheel

Figure 1.2 The Modified Rover wheel

2. Analysis of Manufacturing Constraints and Application of DfX Principles

In order to ensure reliable, first-time-right fabrication of a dual-material off-road tire via Fused Deposition Modeling (FDM) on the Original Prusa XL (0.4 mm nozzle, 0.20 mm layer height), a set of Design for rover wheel according to (DfX) principles and additive manufacturing guidelines must be applied at each stage of the process chain. The following sections justify the selection of key DfX principles and demonstrate their application to our tire design.

2.1 Selected DfX Principles for FDM

1. **Minimum Feature Size & Wall Thickness:**
    - With a 0.4 mm nozzle, features thinner than ≈ 2 mm risk under-extrusion, poor interlayer adhesion, and dimensional inaccuracy.
    - All critical elements (tread lugs, spokes, rim walls) have been maintained at ≥ 2 mm thickness to ensure structural integrity under off-road loads.
2. **Overhang & Self-Supporting Angles:**
    - Overhangs exceeding 45° from the vertical require dense supports and lead to surface defects.
    - The lug flanks and spoke-hub transitions were re-angled to ≤ 45° from vertical, allowing them to print without additional support structures.

3. **Support Minimization & Build Orientation:**
    - Excessive support structures increase print time, material usage, and post-processing effort.
    - By orienting the tire vertically (axis along Z), circumferential features build upon previous layers for natural support, only small internal overhangs in the TPU core require support on build plate.
4. **Part Consolidation & Modularity:**
    - Integrating tread, spokes, and rim into a single CAD model eliminates assembly steps and potential assembly failures.
    - Dual-material zoning is handled within one print job via tool-change commands at the 130 mm diameter boundary.

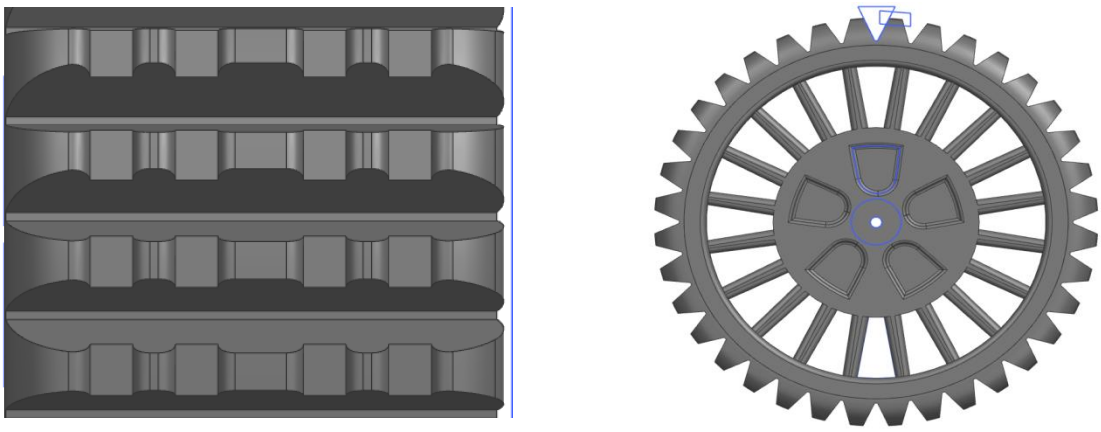


Figure 2.1 Detail of spoke cross-section and block-lug tread showing overhang angles ≤ 45°

2.2 Additive-Manufacturing Process-Chain Guidelines

1. **Pre-Processing (Material & Parameter Selection):**
    - **Materials:** TPU 95A (Thermoplastic polyurethan) for the inner core (130 mm diameter) provides compliance and energy absorption, PETG(Polyethylene Terephthalate Glycol) for the outer tread (130–150 mm) delivers wear resistance.
    - **Temperatures:** Nozzle at 225 °C for TPU and 250 °C for PETG.Bed at 60 °C TPU and 70 °C PETG to optimize adhesion and minimize warping.
2. **Design Phase (Geometry & Infill Strategies):**
    - **Infill:** 10 % gyroid in both materials balances weight reduction, flexibility and print time.
    - **Spoke Geometry:** Twenty radial spokes tapered from 4 mm to 2 mm reduce TPU volume by ≈ 30 % while preserving hub stiffness.
3. **Process Planning (Slicing & Tool-Path Configuration):**
    - **Layer Height:** 0.20 mm for an optimal trade-off between surface finish and build speed.
    - **Shell Count:** Four walls in TPU core and three walls in PETG shell establish the required radial stiffness gradient.
    - **Support Strategy:** Support on build plate within the TPU core only, leveraging self-support in PETG.

- **Tool-Change:** Filament transitions at the 130 mm diamter Z-height, with a small wipe tower to prevent cross-contamination.

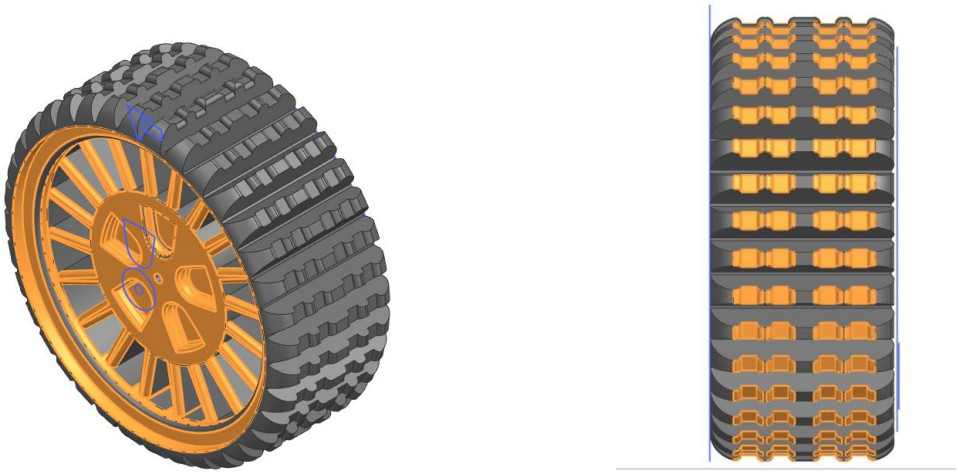
4. **Post-Processing (Support Removal & Surface Finishing):**

- Designed for self-supporting geometry to minimize support volume and simplify cleanup.
- Final inspection of tread edges and spoke roots confirms surface integrity and dimensional accuracy.

3.Optimization of the Component for Manufacturability.

Feature	Original Model	Updated Model	Justification
Spoke Count	Solid disc (0 spokes)	20 radial spokes (4 mm → 2 mm taper)	Reduces material use, improves flexibility on rough terrain
Outer Diameter	150 mm	150 mm	Retains large contact patch for loose surfaces
Rim Diameter	140 mm	130 mm	Aligns with TPU 95A zone, reduces PETG usage
Overall Width	40 mm	60 mm	Distributes load better, minimizes ground sinkage
Tread Thickness	10 mm	20 mm	Improves durability and load resistance
Lug (Block) Depth	3.9 mm	8.6 mm	Enhances traction on soft terrain
Core Zones	Uniform material	Inner 0–130 mm TPU 95A, outer 130–150 mm PETG	Balances shock absorption with wear resistance

Table 1: Comparison of Original and Updated Tire Design Features with JustificationsFigure



3.1:The Modifications done to the original design of the Rover Wheel

3.1 Solutions for Manufacturability

1. **Tread Width Increase:**
    - **Solution:** Expand overall tread width from 40 mm to 60 mm (50 % increase).
    - **Rationale:** A wider contact patch distributes load, reduces ground penetration, and accommodates three PETG perimeters plus gyroid infill for rigidity.
2. **Lug Depth Enhancement:**
    - **Solution:** Increase lug/block depth from 3.9 mm to 8.6 mm.
    - **Rationale:** A deeper lug ensures crisp profile, better mud ejection and stronger layer bonds.
3. **Wall-Thickness Compliance:**
    - **Solution:** Enforce a 2 mm minimum thickness for all spokes, rim walls, and connecting fillets. Spokes are tapered from 4 mm at the hub to 2 mm at mid radius.
    - **Rationale:** ≥ 2 mm ensures reliable extrusion, structural integrity, and controlled compliance where needed.
4. **Dual-Material Zoning:**
    - **Solution:** Partition the tire into an inner core (0-130 mm diameter) for TPU and an outer shell (130-150 mm) for PETG.
    - **Rationale:** TPU core (soft, energy-absorbing) and PETG shell (rigid, wear-resistant) combine complementary properties without mid-air overlaps.
5. **Overhang Angle:**
    - **Solution:** Re-angle all lug flanks and spoke-hub transitions to no more than 45° from vertical.
    - **Rationale:** Self-supporting geometry reduces support volume, print time, and post-processing.

3.2 Off-Road Optimization

- **Traction Enhancement:** Wider and deeper lugs (60 mm × 8.6 mm) to bite into loose substrates.
- **Load Distribution:** Rim diameter reduced from 140 mm to 130 mm, increasing tire width to 60 mm and lowering ground pressure.
- **Compliance Control:** Spokes maintain structural support while allowing radial deflection under load.

3.3 Integration in Siemens NX

1. **Sketch & Feature Updates:**
   - Edited the tread sketch to reflect new width and lug profile.
   - Updated revolve and pattern features to produce deeper lugs with modified flank angles.
2. **Spoke Redesign:**
   - Adjusted the radial spoke command to taper dimensions from 4 mm at the hub to 2 mm at the core boundary.
   - Filleted the hub-spoke junction at 2 mm radius to smooth tool-paths and reduce stress concentrations.
3. **Export for Slicing:**
   - The STL file was exported from Siemens NX as a single component.
   - Imported to PrusaSlicer as a multi-material composite part, preserving the 0–130 mm and 130–150 mm zoning.

4. Creation of the Optimized Component for Additive Manufacturing

4.1 Exporting from Siemens NX

1. **Body Separation**

   In the NX part file,
   - **Inner body (0–130 mm):** TPU core
   - **Outer shell (130–150 mm):** PETG tread
2. **Preview & Validation**
   - The plater view confirms correct color-coded zoning
   - The slice-info panel reports filament usage (TPU: 55.2 m/163.2 g PETG 47.4 m/144.7 g) and total print time (1d2h34 m in Normal mode)

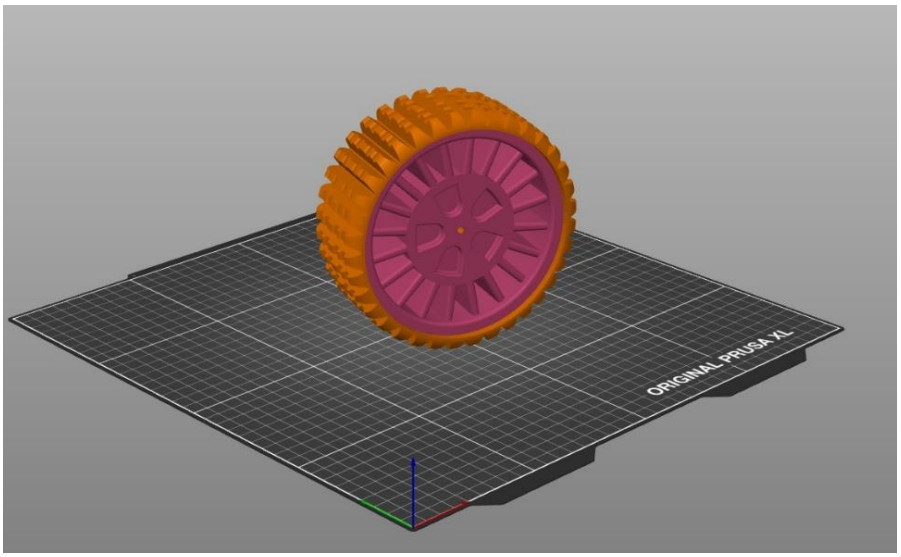


Figure 4.1PrusaSlicer plater view with color-coded TPU vs. PETG regions

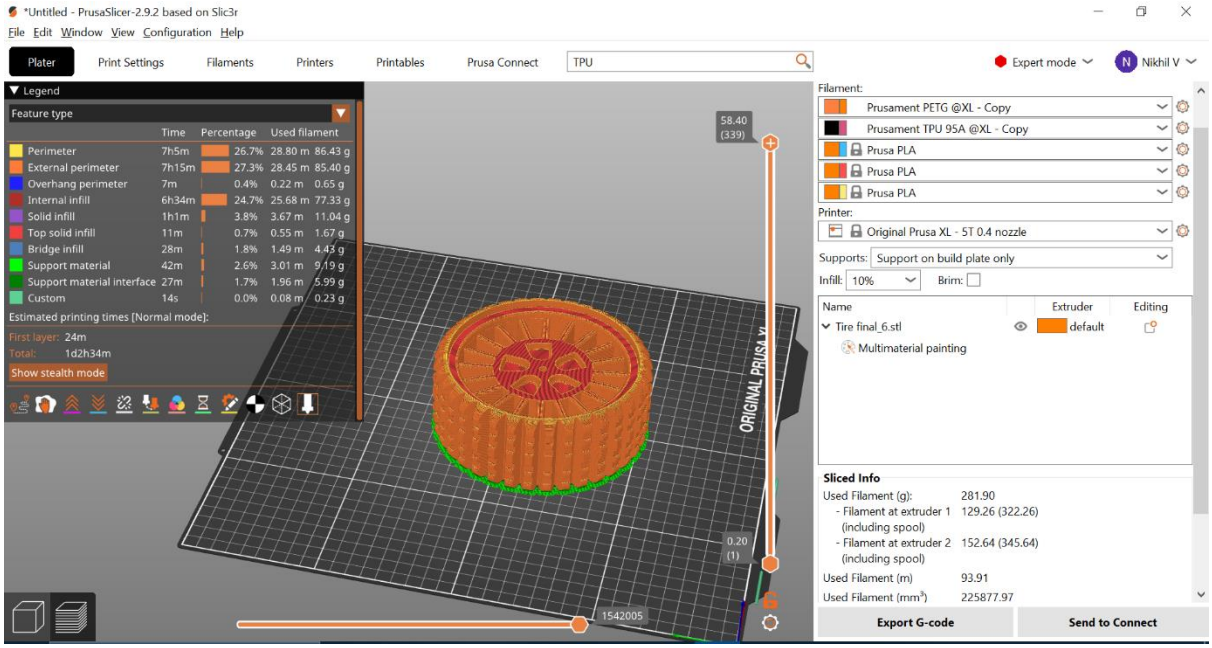


Figure 4.2 Printing in progress, Time durations of various steps

4.2 PrusaSlicer print parameters

Below is a summary of the key print parameters for each material in our dual-extrusion setup:

Parameter	**Outer Shell (PETG) (Extruder 1)**	**Inner Shell (TPU 95A) (Extruder 2)**
Filament Length	28.45m	25.68m
Filament Mass	129.26 g	152.64 g
Infill Density	10%	10%
Infill Pattern	Gyroid	Gyroid
Layer Height	0.20 mm	0.20 mm
Nozzle Temperature	230 °C	240 °C
Bed Temperature	60 °C	70 °C
Wall Count	3	4
Supports	Support on build plate	Support on build plate
Brim	None	None
Print Time (Normal)	1 d 2 h 34 m	
Print Time (Stealth)	1 d 3 h 8 m	
Tool Changes	248	

Table 2 Dual-Material Slicing Parameters for PETG Outer Shell and TPU 95A Inner Shell

5. Conclusion

The rover tire was successfully redesigned for FDM using Siemens NX, adhering to both mechanical and manufacturing requirements. The optimization enhanced off-road capabilities by increasing tread width, depth, and tire surface area. Through integration of DfX principles and slicing in PrusaSlicer, the part is now ready for dual-material FDM printing with improved performance, manufacturability, and structural integrity.

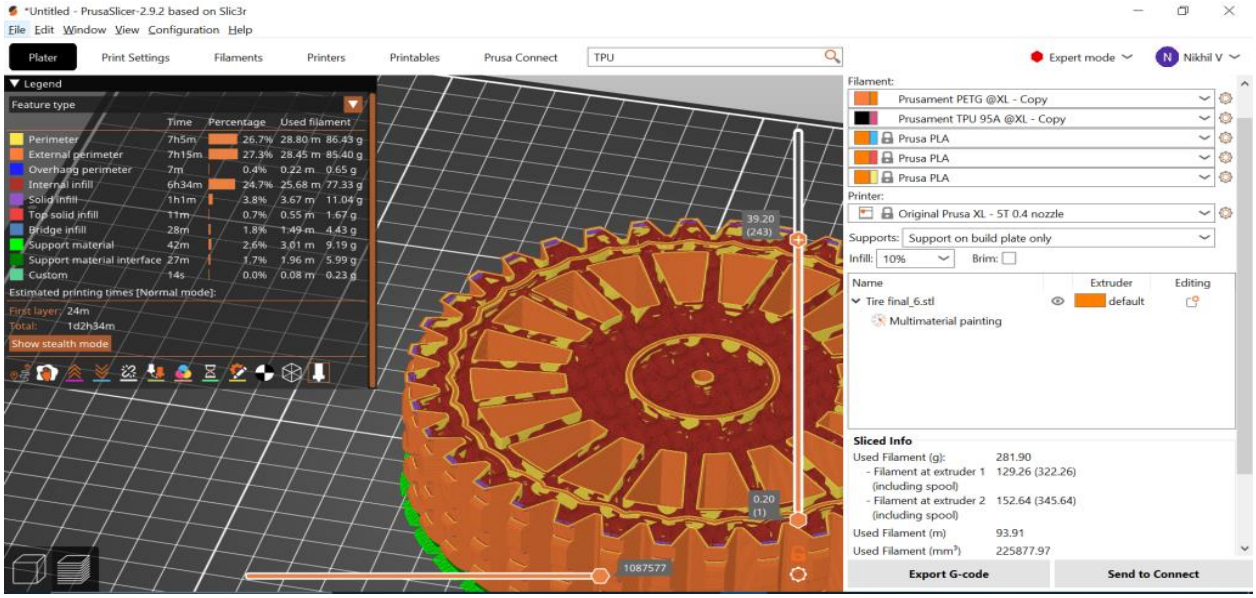


Figure 5.1 Part printing process with time required.

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